

Universal Autonomous AI Workforce Runtime

High-Performance Rust Architecture for End-to-End Remote Digital Employees

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The Paradigm Shift: From Scripted Bots to Autonomous Remote Employees

Traditional AI automation is fundamentally constrained by brittle Python pipelines, single-shot prompt wrappers, and synchronous execution bottlenecks. **Fathom** introduces a production-grade, self-hosted **Rust runtime** designed to instantiate, govern, and orchestrate true **autonomous remote digital employees**. These agents do not merely suggest answers; they independently plan, execute multi-day workflows, operate web browsers via accessibility trees, perform deep OSINT/lead generation, write and maintain software, and interact across all corporate channels 100% remotely.

100% REMOTE OPERATION

Autonomous agents operate independently inside sandboxed environments, interacting with web portals, public registries, APIs, shells, and CRMs 24/7 without manual babysitting or human prompting loops.

MICROSECOND RUST ENGINE

Zero-cost abstractions, Tokio async I/O, and concurrent JoinSet task trees provide sub-millisecond tool dispatch (~0.75 ms), 94 μs memory absorption, and 500x faster cold starts than Python.

UNLIMITED NEURAL COMPUTE

High-throughput routing to cost-efficient frontier foundation models (Kimi k3, Qwen 3.8 Max, GLM 5.3) enables flat-rate monthly seat economics, completely eliminating customer token metering anxiety.

Core Operational Pillars of the Fathom Platform

- **Hierarchical Sub-Agent Swarms:** Recursive coordinator-worker trees with non-blocking broadcast bus telemetry and multi-core CPU parallelism.
- **Autonomous Outreach & OSINT:** 7 unified search engines, corporate website crawlers, pattern-based email inference, and live SMTP port-25 handshake verification.
- **Governed Computer Use:** Native Playwright Chromium integration with Accessibility-Tree snapshots, anti-staleness element caching, and live operator takeover leases.
- **Long-Term Semantic Memory:** Local in-process SQLite engine combining FTS5 BM25 with offline TF-IDF vectors, entity graphs, and sub-5ms volatile prompt digests.
- **Durable Asynchronous Jobs:** SQLite-backed jobs surviving server reboots with self-healing retry task augmentation and automatic failure diagnosis.
- **Enterprise Governance & Trust:** Fail-closed policy evaluation, AES-256-GCM encrypted credentials vault, and immutable audit logs with automated secret redaction.

SYSTEM PERFORMANCE SCORECARD (EMPIRICAL BENCHMARKS ON APPLE SILICON M4)

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RUST WORKSPACE CRATES

0.75 ms

TOOL DISPATCH LATENCY

51+

NATIVE CORE TOOLS

15 MB

DAEMON BASELINE RSS

Strategic Objective: Deliver a scalable, high-frequency trading caliber software foundation where businesses deploy specialized digital workers on demand—scaling operational capacity infinitely without proportional linear headcount expansion.